EDS Lab Assignments

Practical 1

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1.1.1 Calculate Momentum

Problem Statement:

Write a programthataccepts the mass of an object (in kilograms) and its velocity (in meters per second), thencalculates and displays the momentum of the object. The momentum is calculated usingtheformula: p = m * v
where:

m is the mass oftheobject (in kilograms).
v is the velocityoftheobject (in meters per second).

Input Format:

- A singlefloating-point number representing the mass of the object in kilograms.
- A singlefloating-point number representing the velocity of the object in meters per second.

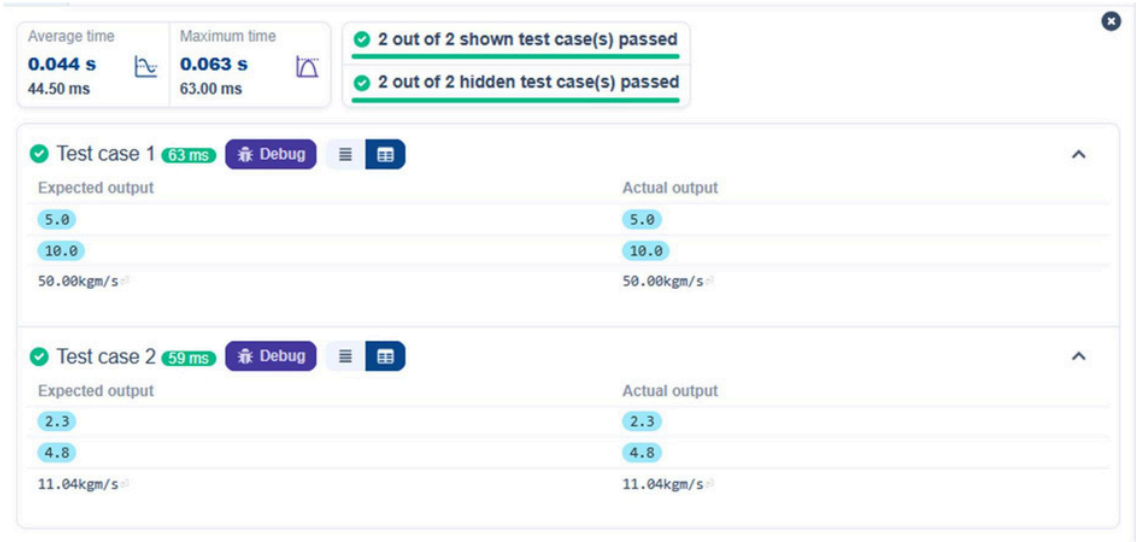
Output Format:

- Theoutputwilldisplay calculated momentum with appropriate units (kgm/s) (roundedupto2decimal places).

Code:

```
mass = float(input())
velocity = float(input())
momentum= mass * velocity
print(f"{momentum:.2f} kgm/s")
```

Output:



1.1.2 Conditional Calculation based on the Number of Digits

Problem Statement:

WriteaPythonprogram that accepts an integer as input. Depending on the number of digits in.

Constraints: 1≤n≤ 999

Input Format:

- The inputconsists of a single integer.

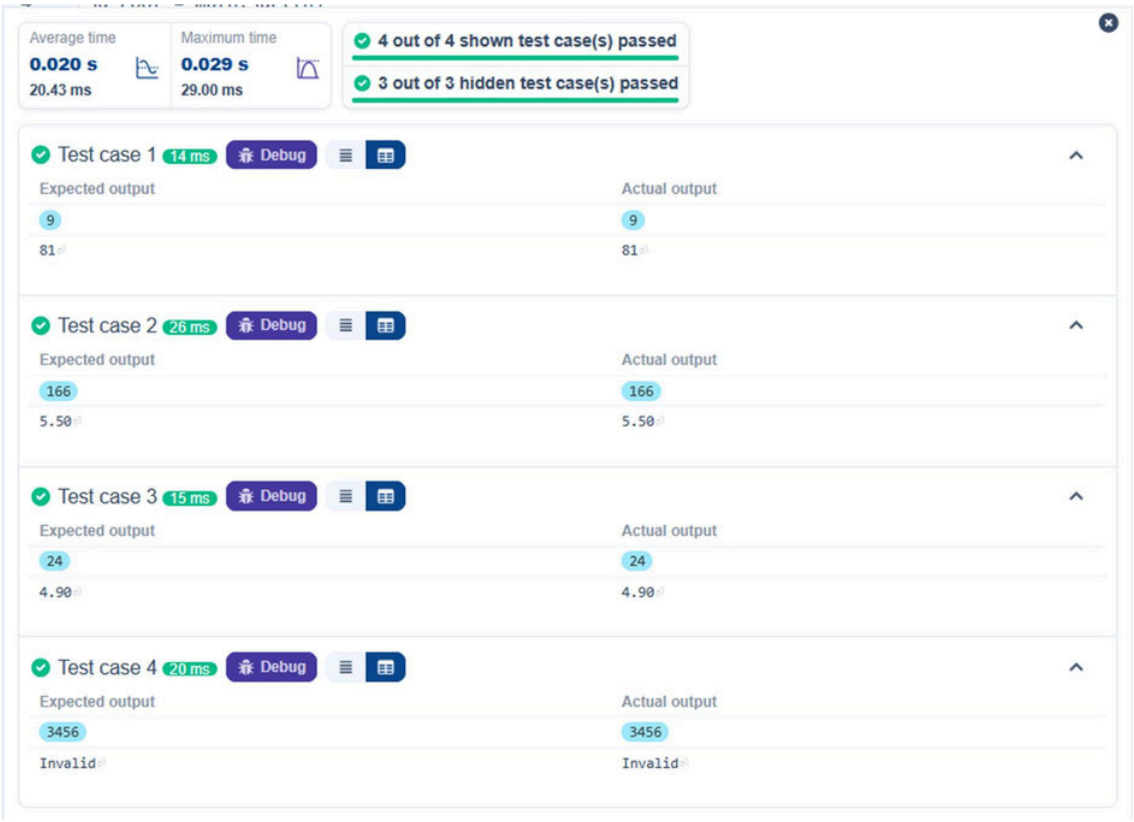
Output Format:

- If is a single-digit number, print its square.
- If is a two-digit number, print its square root (rounded to two decimal places).
- If is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid"

Code:

```
importmath
n=int(input())
sq = n*n
sq_root = math.sqrt(n)
cb_root = math.cbrt(n)
if n>1 and n<=9:
        print(f"{sq}")
elif n>=10 and n<=99:
        print(f"{sq_root:.2f}")
elif n>=100 and n<=999:
        print(f"{cb_root:.2f}")
else:
        print("Invalid")
```

Output:


Average time
0.020 s
20.43 ms
Maximum time
0.029 s
29.00 ms
4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1 14 ms Debug
Expected output
9
81
Actual output
9
81
Test case 2 26 ms Debug
Expected output
166
5.50
Actual output
166
5.50
Test case 3 15 ms Debug
Expected output
24
4.90
Actual output
24
4.90
Test case 4 20 ms Debug
Expected output
3456
Invalid
Actual output
3456
Invalid

1.1.3 Days Between Two Dates

Problem Statement:

Write a Pythonprogram to calculate number of days between two dates.

Input Format:

- The firstline contains the first date in the format YYYY-MM-DD.
- The second line contains the second date in the format YYYY-MM-DD.

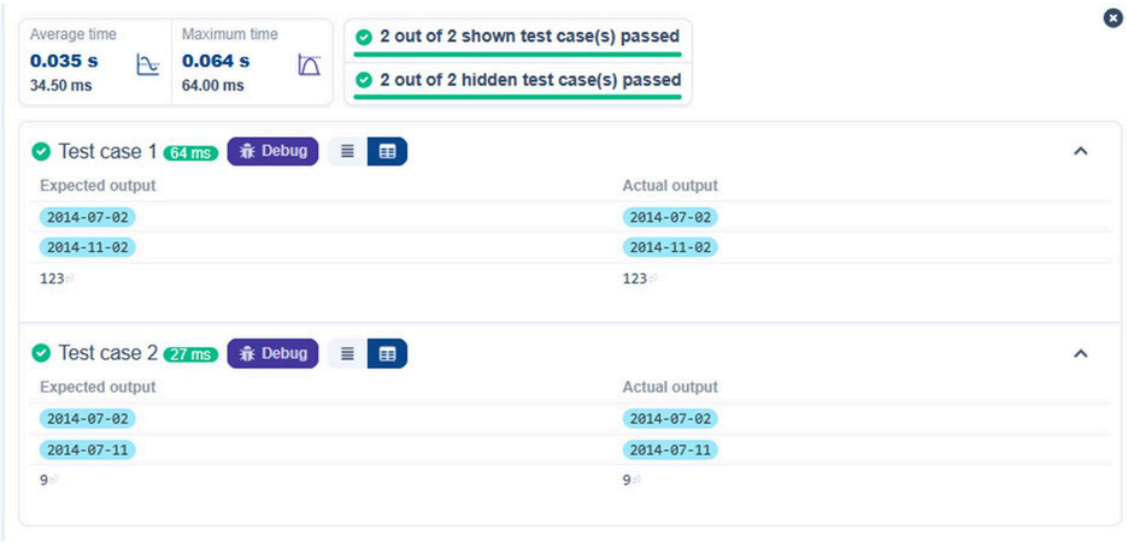
Output Format:

- Anintegerrepresenting the number of days between the given dates.

Code:

```
from datetime import date
from datetime import datetime
date1 = input().strip()
date2 = input().strip()
d1 = datetime.strptime(date1, "%Y-%m-%d") d2 = datetime.strptime(date2, "%Y-%m-%d")
di erence = d2-d1
print(di erence.days)
```

Output:



1.1.4 Reverse a Number

Problem Statement:

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

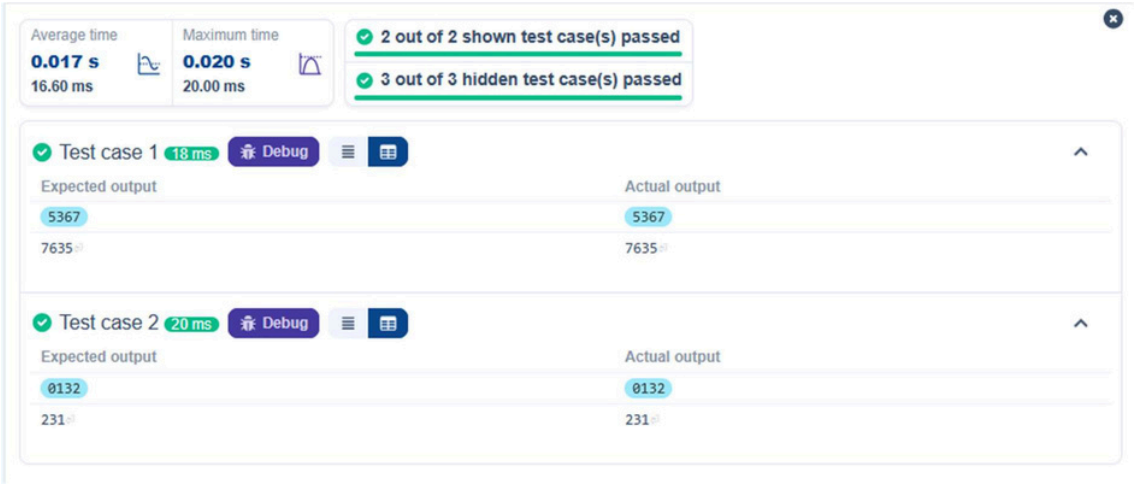
Output Format:

- The output prints a single integer, which is the reversed number.

Code:

```
n=int(input())
reverse=int(str(n)[::-1])
print(reverse)
```

Output:



1.1.5 Multiplication Table

Problem Statement:

Write a Pythonprogram that takes an integer as input and prints the multiplication table forthat integerfrom 1 to 10.

Input Format:

- The firstline of input contains an integer that represents the number for which the multiplication table is to be printed.

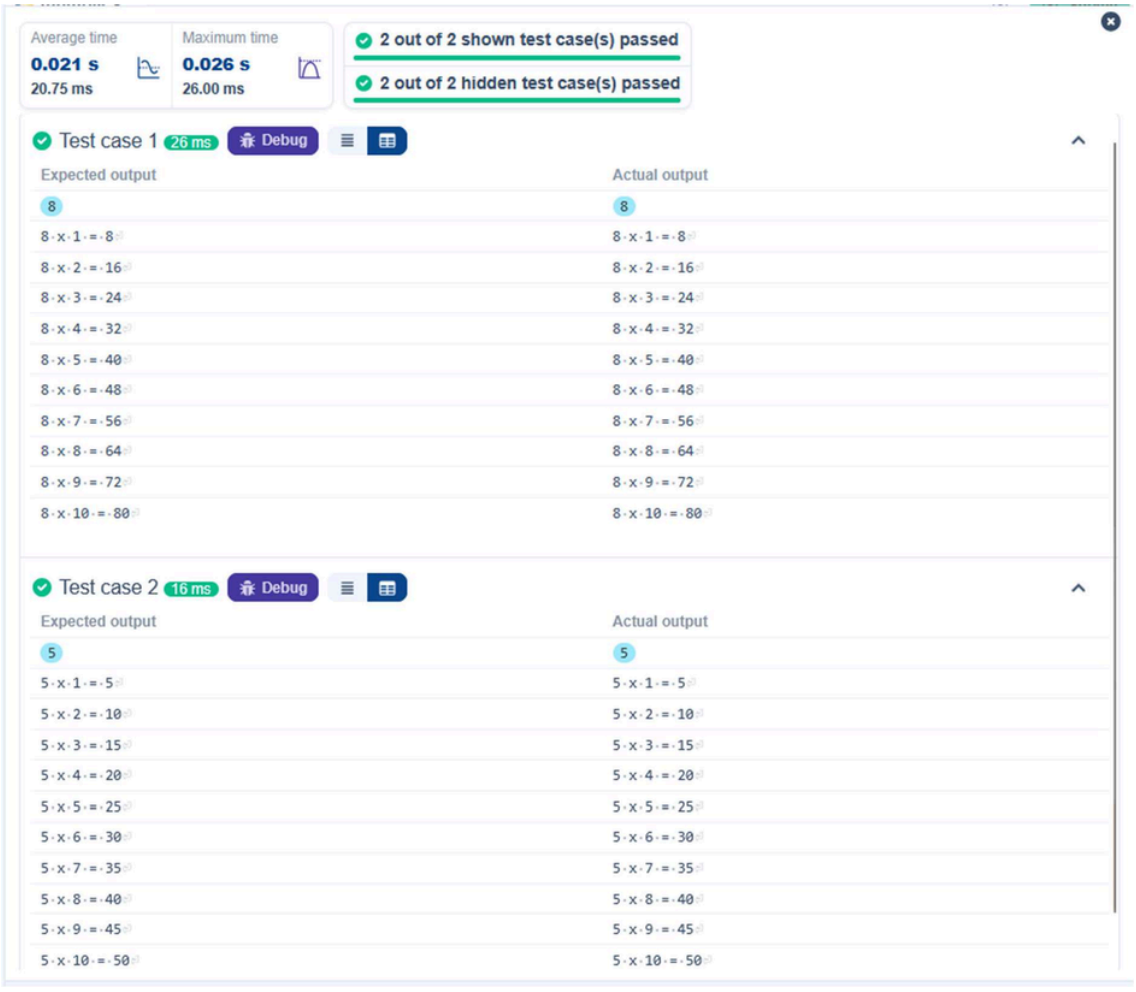
Output Format:

- Printthemultiplication table for the given number in the format: <number> x <multiplier> = <result>

Code:

```
n = int(input())
for i in range(1,11):
    print(n, "x", i, "=", n*i)
```

Output:



1.2.1 Pass or Fail

Problem Statement:.
Writea Pythonprogram that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade isFirst Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade isSecond Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade isThird Division.

Input Format:

- The firstinput will be an integer, the number of courses.
- The second input will be integers representing the marks of the student in each of the courses, separated by a space.

Output Format:
Ifthestudent passes all courses:

- Print theaggregate percentage (formatted to two decimal places).
- Print thegrade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

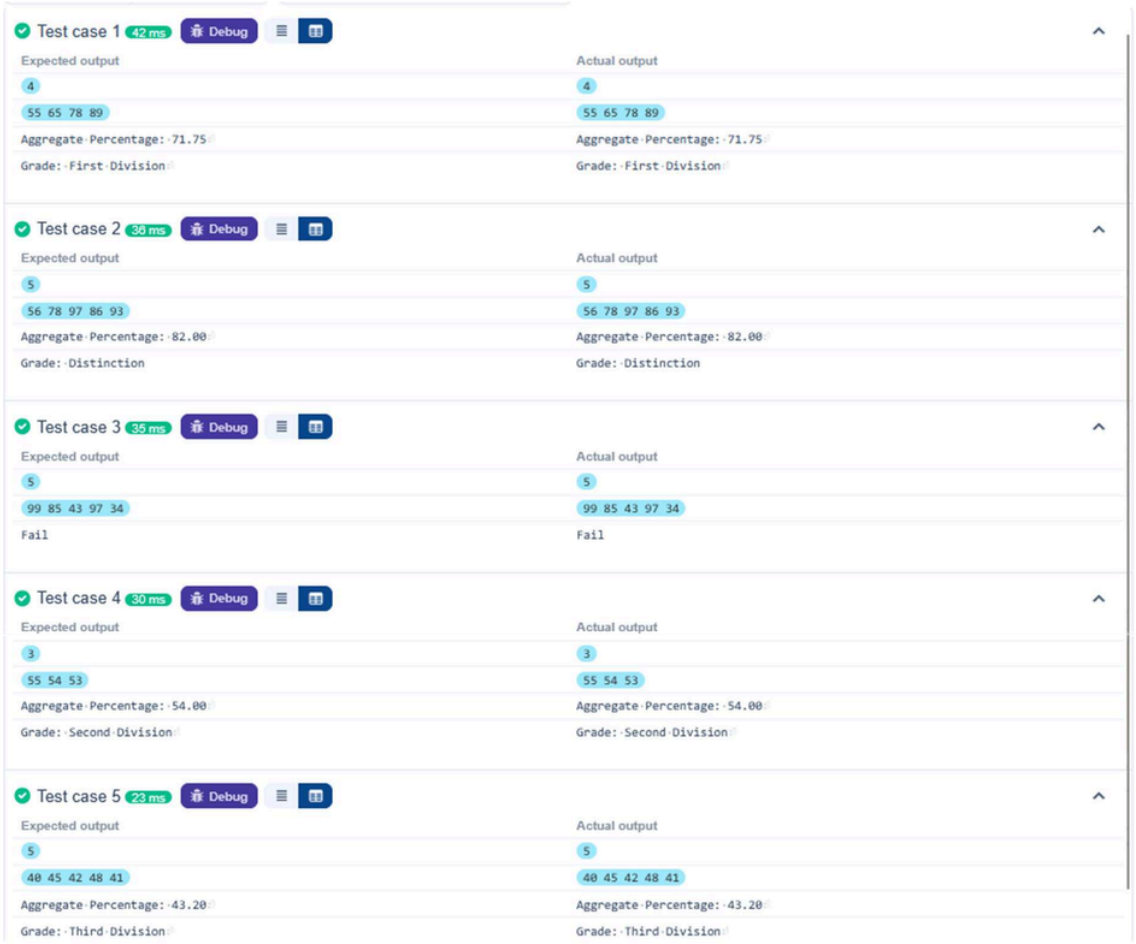
- "Fail".

Code:

```
defcalculate(marks, total_courses):
        if any(mark<40 for mark in marks):
                return "Fail"
        total_marks = sum(marks)
        aggregate_percentage = ((total_marks)/(total_courses*100))*100
        if aggregate_percentage > 75:
                grade = "Distinction"
        elif aggregate_percentage >= 60:
                grade = "First Division"
        elif aggregate_percentage >= 50:
                grade = "Second Division"
        elif aggregate_percentage >= 40:
                grade = "Third Division"
        return (aggregate_percentage, grade)
num_courses = int(input())
marks = list(map(int,input().split()))
result = calculate(marks, num_courses)
if result == 'Fail':
        print("Fail")
else:
```

```
aggregate_percentage, grade = result
print("Aggregate Percentage:",f"{aggregate_percentage:.2f}")
print(f"Grade: {grade}")
```

Output:



1.2.2 Fibonacci Series

Problem Statement:
Write a Python program that uses recursion to print the first terms of the Fibonacci series.

Input Format:

- Asingleinteger representing the number of terms to generate.

Output Format:

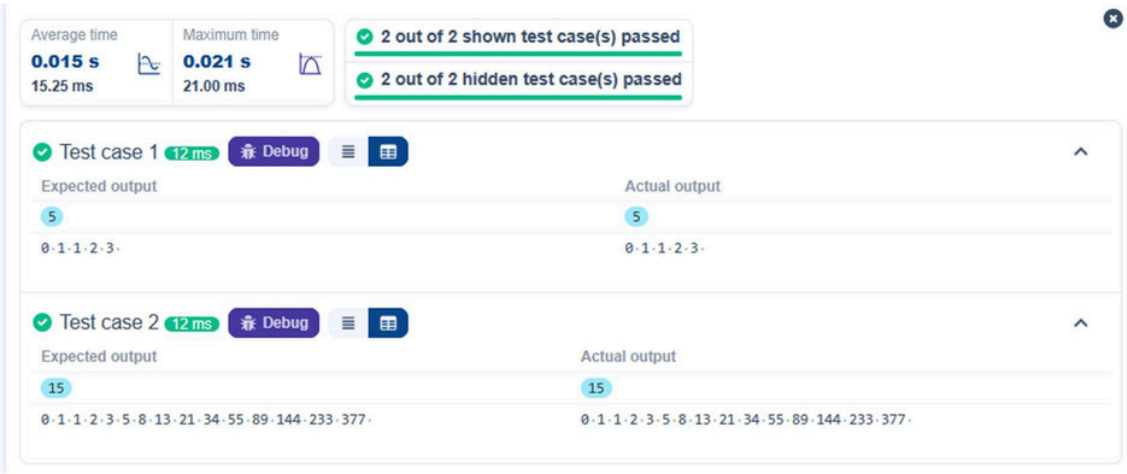
- Asingle line containing the first terms of the Fibonacci sequence, separated by spaces.

Code:

```
deffibonacci(n, a=0, b=1):
        if n==0:
                return a
        else:
                return fibonacci(n-1,b,a+b)

n=int(input())
for i in range(n):
        print(fibonacci(i) ,end=" ")
```

Output:



1.2.3 Pattern-1

Problem Statement:
Writea Pythonprogram to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- Theinputis an integer, representing the number of rows in the pattern.

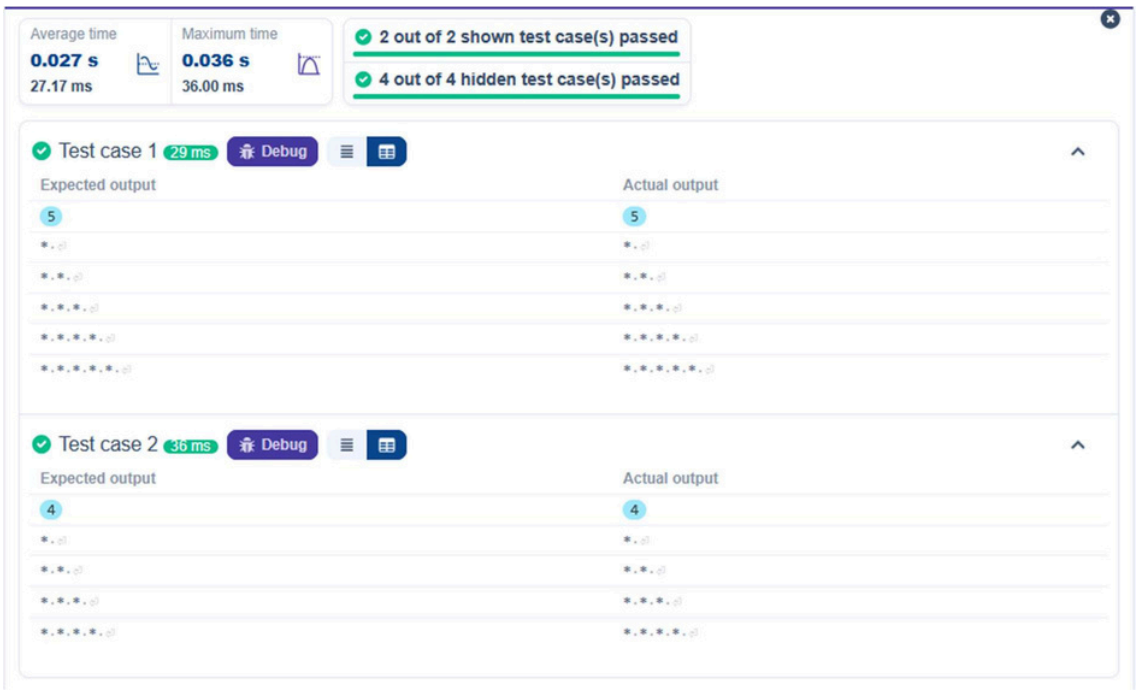
Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Code:

```
rows =int(input())
for i inrange(1, rows+1):
        for j in range(i):
                print("* ",end = "")
        print()
```

Output:



1.2.4 Pattern-2

Problem Statement:

Write a Pythonprogram to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Code:

```
n=int(input())
for i in range(1, n+1):
        for j in range(1, i+1):
                print(j,end=" ")
        print()
```

Output: